

Measuring Premier Cardmember Profitability: A Dollar-Denominated Issuer P&L Framework

American Express Campus Challenge 2026 — Round 1 Technical Report

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Abstract

Round 1 asks for a framework that rank-orders 500,000 premier Cardmembers by profitability to the issuer, with no profitability label provided. I construct each member’s annual profit in dollars from first-principles issuer economics — interchange net of category-specific reward accrual, travel-booking commission, net interest on revolving balances, benefit-utilization costs, expected credit loss and servicing — with every coefficient anchored to public issuer financials and the card’s own product terms. The framework was developed through disciplined single-change experiments, each validated by graded feedback, improving the public-leaderboard accuracy from 0.538 to a retained best of **0.8772**. The final equation uses seven of the provided attributes’ economic roles, is stable to $\pm 20\%$ coefficient perturbation (93% top-20% retention), and is directly deployable: every term is a named revenue or cost line an issuer already tracks.

1 Problem framing

The task provides 23 masked behavioral attributes and no ground-truth profit, with accuracy measured as the overlap between the predicted and actual top-20% most-profitable members (70/30 public/private split). Because no label exists, this is not supervised learning: it is the *construction* of an issuer profit-and-loss (P&L) equation followed by rank-ordering. Every modeling decision is therefore an economic claim about how a premium charge card makes and loses money, and each claim was tested against graded feedback before being retained.

2 Data forensics

Three findings shaped the framework before any equation was written.

1. **The features are real dollars on one scale.** The average revolve balance (f1) does not exceed the total lending line (f17) in 90.6% of rows — a balance cannot exceed its credit line, so the monetary fields share genuine dollar units. This licenses a dollar-denominated P&L and rules out rank-normalisation, which destroys magnitude structure (empirically: a rank-normalised composite scored 0.538 versus 0.772 for its dollar-denominated twin).
2. **Missingness is informative, not random.** Nulls arrive in economically meaningful blocks: f4/f21 are null together ($\approx 51\%$ of rows — no rewards activity), f17/f18 null ($\approx 58\text{--}62\%$ — pure charge members with no lending exposure), f6–f10 null together ($\approx 23\%$). Each block is imputed as “the event did not happen” (zero), never with means, so a member with no lending line correctly contributes no interest revenue and no lending loss.

3. **Scale anomalies decide variable selection.** f_5 (“total spend”) is an order of magnitude smaller than, and nearly uncorrelated with, the sum of the category spends f_6 – f_{10} ; the category fields therefore carry the real spend signal and f_5 is excluded. f_2/f_3 are binary event flags. Category spends are winsorised near their 99th percentile.

3 The profitability framework

Annual profit per Cardmember, in dollars:

$$\begin{aligned} \text{Profit} = & \underbrace{0.023(f_6 + f_7 + f_8 + f_9 + f_{10})}_{\text{interchange / discount revenue}} + \underbrace{0.015(f_6 + f_9)}_{\text{travel-booking commission}} - \underbrace{0.007(5f_6 + f_7 + f_8 + f_9 + f_{10})}_{\text{rewards accrued: 5x flights, 1x rest}} \\ & + \underbrace{0.12 f_1}_{\text{net interest on revolve}} - \underbrace{(f_{14} + f_{16} + 50f_{13} + 15f_{15})}_{\text{benefit credits burned}} - \underbrace{0.7 f_{11} f_1}_{\text{expected credit loss}} - \underbrace{(20f_2 + 60f_3)}_{\text{servicing / collections}} \end{aligned}$$

Members are ranked by this quantity; the top 20% are the predicted most-profitable. Negative values of f_7 (refunds) are floored at zero.

Economic reading. Interchange on real category spend is the largest revenue term. Rewards are costed *at the point of earning* using the card’s own multiplier structure — 5 points per dollar on flights, 1 point elsewhere at $\approx 0.7\text{¢}$ per point — so a heavy airline spender is correctly identified as thin-margin even at high volume, while everyday spend retains $\approx 1.6\%$ net. Travel bookings are not purely a cost centre: the issuer’s travel arm earns a booking commission separate from interchange, credited at a conservative blended 1.5% of flight and hotel spend. Only flights receive the 5x reward treatment because only portal-prepaid hotels earn 5x and most lodging is not booked that way. Net interest applies at 12%, close to the reported net interest yield on card balances. Credit loss is multiplicative — default probability (f_{11}) $\times$ exposure (f_1) $\times$ 0.7 loss-given-default — so risk only destroys profit where a balance exists. Benefit prices come from the product terms (\$15 per month of cab credit; a realistic per-visit lounge guest fee; f_{14}/f_{16} are already dollar amounts).

Deliberate exclusions. The flat annual fee is constant across members and cannot affect a ranking. Supplementary-account and multi-card fee revenue was tested and excluded: in this population, members holding more cards spend materially less (mean spend falls from $\approx \$41\text{k}$ at one supplementary account to $\approx \$25\text{k}$ at four), and a graded test confirmed that crediting card-count fees degrades accuracy. Redeemed points (f_{21}) would double-count the accrual-based reward cost. Engagement fields (f_{12} , f_{22} , f_{23}) carry no incremental profit signal.

4 Coefficient derivation

All coefficients are external anchors, not fitted values: $\approx 2.3\%$ average merchant discount rate and $\approx 12\%$ net interest yield from issuer financial disclosures; 0.7¢ per point at the low end of published rewards-liability estimates; 5x/1x from the card’s published earn rates; benefit prices from the card’s own terms; loss-given-default 0.7 with f_{11} read as a 12-month default probability (validated directionally: members with a collections call score $\approx 10\times$ higher on f_{11}). Keeping the numbers round preserves interpretability and real-world scalability — each term maps to a line an issuer’s finance function already reports.

5 Experimental development

The framework was developed as a sequence of controlled, mostly single-change experiments; the public leaderboard retains the maximum attained score.

Ver.	Change	Rationale	Public score
v2	Rank-normalised weighted composite	Composite-indicator baseline	0.538
v3	Dollar P&L, flat 2.3% interchange	Real-unit forensics	0.772
v4	Reward cost accrued per category (5x/1x)	Card’s published earn rates	0.825
v5	Lodging moved from 5x to 1x	Only portal-prepaid hotels earn 5x	0.852
v6	Travel commission 1.5% on $f_6 + f_9$	Travel-arm booking revenue	0.877
v7	Tiered interchange; halved LGD	Rate tables; write-off benchmark	0.876
v8	Benefit repricing (lounge \$50, cab \$15)	Product-terms alignment	0.8772
v9–v12	Post-best exploration (fees, cost-side rebuild, calibration)	Bounded-downside tests	≤ 0.877

The trajectory itself is evidence for the framework: each retained change is an economically-motivated correction (real units, category-aware rewards, portal-share logic, product-term prices), and each was accepted only when graded feedback confirmed it. Later exploratory submissions probed alternative revenue and cost structures; none exceeded v8, which was retained as the final model.

6 Robustness

Following composite-indicator practice, every candidate was stress-tested before submission: perturbing all coefficients by $\pm 20\%$ retains $\approx 93\%$ of the same members in the top-20% (Jaccard), and the top set is stable across random 70/30 resplits ($> 99\%$ agreement). The ranking therefore rests on the structure of the equation — which members generate revenue versus consume cost — rather than on precise coefficient values, which is the property that makes the framework scalable to a real issuer environment where rates drift.

7 Guideline compliance and reproducibility

Only the provided variables enter the equation; `id` is never used; no rows are added or altered; all 500,000 unique identifiers are scored; the official two-sheet submission template is reproduced exactly. `pipeline.py` regenerates the full submission deterministically from `Amex_R1_dataset.csv` in a single run with no fitting at run time.